

METTL3-mediated m6A modification of MT1G inhibits papillary thyroid carcinoma cell growth and metastasis via Wnt/ β -catenin pathway

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ABSTRACT

Background: Downregulation of metallothionein 1 G (MT1G) has been demonstrated in papillary thyroid carcinoma (PTC) tissues. However, the underlying molecular mechanisms of MT1G in PTC progression need to be further explored.

Methods: MT1G and methyltransferase-like 3 (METTL3) mRNA levels were tested by quantitative real-time PCR. The protein levels of MT1G, METTL3, Wnt3A and β -catenin were measured by western blot. Cell proliferation, apoptosis, invasion and migration were measured by cell counting kit 8 assay, colony formation assay, EdU assay, flow cytometry, transwell assay and wound healing assay. MeRIP analysis was used to detect the MT1G methylation. The interaction between METTL3 and MT1G was evaluated using RIP assay and dual-luciferase reporter assay. A mouse xenograft model was also constructed to explore the roles of METTL3 and MT1G *in vivo*.

Results: MT1G expression was downregulated in PTC, and its overexpression suppressed PTC cell growth, invasion and migration. METTL3-regulated m6A modification enhanced MT1G mRNA stability. Overexpression of METTL3 repressed PTC cell growth and metastasis, and this effect was reversed by MT1G knockdown. Besides, METTL3/MT1G axis could inhibit the activity of Wnt/ β -catenin pathway. Meanwhile, METTL3 enhanced MT1G expression to suppress PTC tumor growth through Wnt/ β -catenin pathway *in vivo*.

Conclusion: METTL3-mediated m6A modification of MT1G inhibited PTC cell growth and metastasis via inactivating the Wnt/ β -catenin pathway.

1. Introduction

Papillary thyroid carcinoma (PTC) is the most common type of thyroid cancer, accounting for about 90 % of cases (Papaleontiou et al., 2021; Melik et al., 2022). PTC is considered an inert tumor with a 10-year survival rate of 93 %, and its incidence has gradually increased in recent years (McLeod et al., 2019; Clark, 2011). The vast majority of PTC can be completely resected by surgery, and the majority of patients have a good prognosis with postoperative drug therapy (Paluskievicz et al., 2022). Despite the low rate of recurrence or metastasis in most PTCs, a small proportion of patients exhibit tumor heterogeneity with more aggressive variants (Coca-Pelaz et al., 2020). The development of targeted therapy has brought new hope to PTC patients (Yang et al., 2023). Therefore, our study aims to reveal the molecular mechanism and potential targets of PTC.

Metallothioneins (MTs), a class of cysteine-rich small molecule proteins (Wang et al., 2019; Xu et al., 2022), play a key role in tumorigenesis (Si and Lang, 2018). MT1G, belonging to the MT1 subtype, has

been confirmed to have abnormal expression in a wide variety of tumors, such as colorectal cancer (Peng et al., 2022) and hepatocellular carcinoma (Wei et al., 2022a). Importantly, it had been demonstrated that MT1G was lowly expressed in PTC and exerted anti-cancer effects in PTC (Ferrario et al., 2008). Consistent with this results, we used GEO database (GSE165724) to analyze the differentially expressed genes and discovered that MT1G expression was significantly reduced in PTC tissues. Therefore, MT1G may be a regulator for PTC progression, and its role and molecular mechanism in PTC need to be further investigated.

The development of PTC is closely related to epigenetic aberrations (Mastronikolis et al., 2020). N6-methyladenosine (m6A) is a common important epigenetic alteration (Zhu et al., 2023), and it-associated proteins include methyltransferases, demethylation enzymes, and methyl-binding proteins (Liu et al., 2018). Methyltransferase-like 3 (METTL3), an m6A “writer” proteins, is the only catalytic subunit (Zeng et al., 2020; Wang et al., 2016). Recent studies have demonstrated that METLL3 plays important roles in a variety of cancers, which is mostly dependent on its m6A methyltransferase activity (Li et al., 2019). Hou

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et al. showed that METTL3 was lowly expressed in PTC tissues (Hou et al., 2020). Besides, METTL3-mediated m6A/c-Rel/IL-8 axis inhibited the progression of PTC (He et al., 2021). Importantly, the low expression of MT1G in PTC had been confirmed to be related to hypermethylation (Huang et al., 2003). In this, we found that MT1G had m6A modification sites and could interact with METTL3. However, whether METTL3 regulates PTC progression by mediating MT1G expression through m6A modification has not been reported.

Here, we demonstrated that METTL3 inhibited PTC development by mediating the m6A modification of MT1G. The findings can provide a new direction for the clinical treatment of PTC.

2. Materials and methods

2.1. Samples

Our research had been approved by the Ethics Committee of Cangzhou Hospital of Integrated TCM-WM and all operation procedures comply with relevant regulations. After participant signed written informed consent, PTC tissues (n = 31) and normal thyroid tissues (n = 31) were collected from 31 PTC patients at Cangzhou Hospital of Integrated TCM-WM. All samples were preserved in -80°C for subsequent studies.

2.2. Cell culture and transfection

PTC cells (TPC-1, SW579 and SW1736) and Nthy-ori3-1 epithelial cells were purchased from Biovector NTCC (Beijing, China). They were cultured in RPMI-1640 or DMEM containing 1 % penicillin-streptomycin and 10 % FBS (Gibco, Grand Island, Carlsbad, USA) at 37°C with 5 % CO_2 . PTC cells were transfected with pcDNA overexpression vectors or shRNAs targeting METTL3/MT1G (sh-METTL3/sh-MT1G), and corresponding controls (RiboBio, Guangzhou, China) using Lipofectamine 3000 (Invitrogen, Carlsbad, CA, USA).

2.3. Quantitative real-time PCR (qRT-PCR)

Total RNAs were extracted by TRIzol reagent (Invitrogen) and then used for synthesizing cDNA by PrimeScript RT Reagent Kit (Takara, Tokyo, Japan). After that, cDNA was mixed with SYBR Green (Takara) and specific primers (Table 1) for PCR amplification. Relative mRNA expression was calculated by $2^{-\Delta\Delta\text{Ct}}$ method with GAPDH as internal control.

2.4. Western blot (WB)

TPC-1 and SW579 cells were lysed with RIPA buffer (Beyotime, Shanghai, China) for extracting total proteins. Extracted proteins were separated by SDS-PAGE and transferred to PVDF membranes. Membrane was closed with blocking buffer (Beyotime), and incubated with anti-MT1G (#OM263051, 1:1000, Omnimaba, Alhambra, CA, USA), anti-METTL3 (ab195352, 1:1000, Abcam, Cambridge, MA, USA), anti-Wnt3A (ab219412, 1:1000), anti- β -catenin (ab8227, 1:1000), anti-GAPDH (ab9485, 1:2500) and secondary antibody (ab205719/ab205718, 1:10000/1:50000). Afterwards, membrane was incubated

with BeyoECL Plus solution (Beyotime) to detect protein signals.

2.5. Cell counting kit 8 (CCK8) assay

TPC-1 and SW579 cells were seeded in 96-well plates and incubated with CCK8 reagent (Beyotime) for 2 h after cultured for 48 h. Cell viability was tested by a microplate reader at 450 nm.

2.6. Colony formation assay

TPC-1 and SW579 cells were seeded in 6-well plates and cultured for 14 days. After that, colonies were fixed with 4 % paraformaldehyde, washed with PBS and stained with crystal violet (0.1 %; Beyotime). The formed colonies were visualized and then counted under a microscope.

2.7. EdU assay

According to EdU Cell Proliferation Kit (Beyotime) instructions, TPC-1 and SW579 cells were seeded in 6-well plates, labelled with EdU solution, fixed with 4 % paraformaldehyde and permeated with 0.3 % TritonX-100. Then, PTC cells were incubated with DAPI solution, and EdU positive cell rate was analyzed under a fluorescence microscope.

2.8. Flow cytometry

TPC-1 and SW579 cells were collected and resuspended with binding buffer. Then, cells were stained with Annexin V-FITC and PI (Sigma-Aldrich, St. Louis, MO, USA). Flow cytometry was used to analyze apoptosis rate.

2.9. Transwell assay

Diluted Matrigel was spread on the upper of transwell chambers (BD Biosciences, San Jose, CA, USA) and cultured for 3 h, while the lower chamber was filled with complete medium. TPC-1 and SW579 cells suspended with serum-free medium were seeded in the upper chamber for culturing 24 h. Afterwards, the invaded cells were stained with crystal violet solution. Invasive cell number was counted under a microscope.

2.10. Wound healing assay

TPC-1 and SW579 cells were inoculated in 24-well plates and cultured until 90 % confluences. Using the ruler as a reference, a 10 μL pipette was used to create a wound in cell layer, and then cell was photographed as 0 h. After cultured with serum-free medium for 24 h, migration distance was calculated under a microscope.

2.11. MeRIP assay

MeRIP assay was performed according to the m6A MeRIP Kit (Cloud-seq, Shanghai, China) instructions. In brief, RNA was fragmented after being divided into the input and IP groups. IP group was incubated with m6A antibody, IgG antibody and PGM Beads for 1 h. After that, the supernatant was discarded, RNA was purified, and the enrichments of MT1G and RARA (positive control) were detected by qRT-PCR with specific primers.

2.12. RIP assay

The binding ability of METTL3 to MT1G protein was assayed by RIP Kit (Sigma-Aldrich). Cells transfected with sh-NC/sh-METTL3 were lysed with RIP lysis buffer, and the lysates were incubated with anti-METTL3 and magnetic beads. The magnetic bead-antibody conjugates were centrifuged and unbound material was washed away after resuspension with RIP lysis buffer. The magnetic bead-antibody conjugate

Table 1

Primer sequences used for qRT-PCR.

Name		Primers for PCR (5'-3')
METTL3	Forward	CAGAGGCAGCATGTCTCCA
	Reverse	ATGGACACAGCATCAGTGGG
MT1G	Forward	TTCCCTTCTCGCTTGGGAAC
	Reverse	TTTGTACTTGGGAGCAGGGC
GAPDH	Forward	GGAGCGAGATCCCTCCAAAT
	Reverse	GGCTGTTGTCATCTTCTCATGG

was treated with TRIzol reagent to isolate precipitated RNA and then qRT-PCR was performed to measure MT1G enrichment.

2.13. Dual-luciferase reporter assay

MT1G regions containing wild-type and mutant m6A motifs were cloned into pGL3-basic vectors to generate wild-type MT1G/MUT-MT1G vectors. TPC-1 and SW579 cells were co-transfected with pcDNA/METTL3 overexpression vector and wild-type MT1G/MUT-MT1G vectors. Then, cells were incubated with Dual Luciferase Reporter Gene Assay Kit II (Beyotime) to detect luciferase activity.

2.14. Actinomycin D (Act D) assay

TPC-1 and SW579 cells transfected with sh-NC/sh-METTL3/pcDNA/METTL3 overexpression vector were treated with Act D (MedChemExpress, Monmouth Junction, NJ, USA) for 0, 3, 6 and 9 h. At each time point, cells were collected, and MT1G mRNA expression was detected according to the qRT-PCR procedure.

2.15. Mice xenograft models

BALB/c nude mice (Vital River, Beijing, China) were subcutaneously injected with TPC-1 cells co-transfected with lentivirus pcDNA, lentivirus METTL3 overexpression vector, sh-NC or sh-MT1G in the flank area ($n = 6$). Tumor volumes and weights were measured starting at 7 days, and then every 3 days thereafter. Tumor tissues were collected after 22 days. Additionally, tumor tissues were lysed for WB analysis and prepared as paraffin sections to perform immunohistochemical (IHC) staining. Animal studies were approved by the Animal Ethics Committee of Cangzhou Hospital of Integrated TCM-WM.

2.16. IHC staining

Paraffin sections of tumor tissues were prepared and incubated with anti-Ki67 (ab15580, 1:200), anti-MMP9 (ab76003, 1:100) and anti-MT1G (1:100, Omnimab) according to SP Kit (Solarbio, Beijing, China) instructions. The positive cells of Ki67, USP7 and MT1G were observed under a microscope.

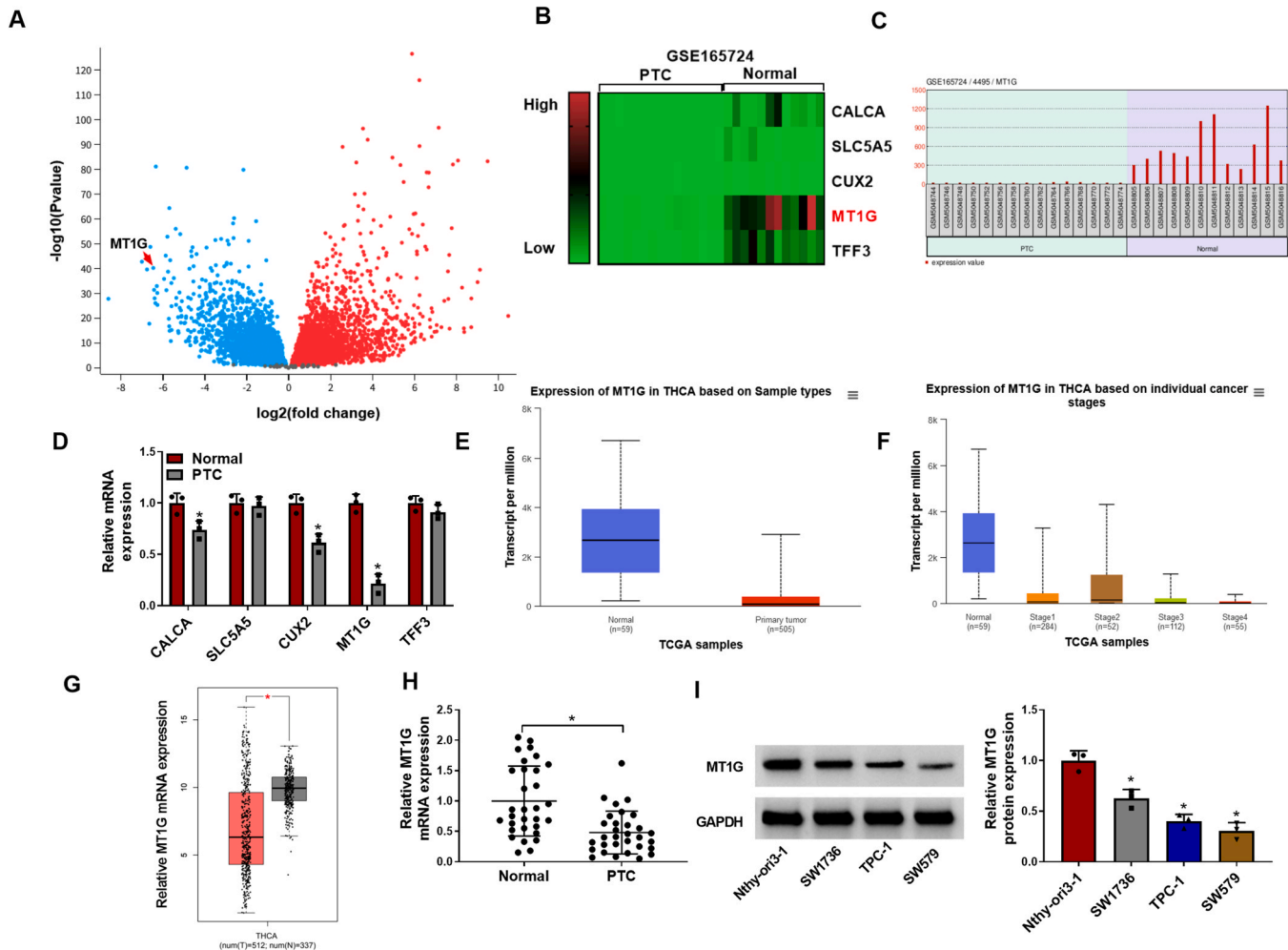


Fig. 1. Expression of MT1G in PTC tissues and cells. (A) Volcano plot shows differentially expressed genes between PTC tissues ($n = 15$) and normal thyroid tissues ($n = 12$) in the GEO database. (B) GEO database analysis showed the levels of CALCA, SLC5A5, CUX2, MT1G and TFF3 in PTC tissues ($n = 15$) and normal thyroid tissues ($n = 12$). (C) Levels of CALCA, SLC5A5, CUX2, MT1G and TFF3 were tested by qRT-PCR in PTC tissues and normal thyroid tissues ($n = 3$). (D-F) TCGA and GEPIA database were used to predict the MT1G expression in PTC tissues and normal thyroid tissues. (G) qRT-PCR was performed to analyze MT1G mRNA levels in normal thyroid tissues ($n = 31$) and PTC tissue ($n = 31$). (H) MT1G protein levels in PTC cells and Nthy-ori3-1 cells were measured by WB analysis ($n = 3$). * $P < 0.05$.

2.17. Statistical analysis

Data are presented as mean $\pm$ SD using GraphPad Prism 7.0 software. Results were compared between different groups using Student's *t*-test or ANOVA. $P < 0.05$ was considered as significant.

3. Results

3.1. MT1G expression was decreased in PTC

Analysis of GEO database (GSE165724) identified the top 5 differentially expressed genes in PTC tissues and normal thyroid tissues, of which MT1G was markedly downregulated in PTC tissues (Fig. 1A-B). The qRT-PCR assay also showed the same results (Fig. 1C). Based on this, we aimed to explore the role of MT1G in PTC. The TCGA database predicted that MT1G was underexpressed in PTC tissues compared with that in normal samples, and its expression was correlated with tumor stage (Fig. 1D-E). GEPIA database also confirmed the downregulation of MT1G mRNA expression in PTC tissues (Fig. 1F). To further confirm this, we measured MT1G expression in 31 paired PTC tissues and normal tissues, and found that the mRNA expression of MT1G in PTC tissues was significantly downregulated (Fig. 1G). In addition, MT1G protein levels were decreased in PTC cells (Fig. 1H).

3.2. Overexpression of MT1G inhibited PTC cell growth and metastasis

To study the role of MT1G in PTC process, SW579 and TPC-1 cells were transfected with MT1G overexpression vector. The increased MT1G protein expression indicated that the transfection was successful (Fig. 2A). MT1G overexpression suppressed viability, colony numbers and EdU positive cell rate, while promoted apoptosis in SW579 and TPC-1 cells (Fig. 2B-F). Furthermore, MT1G overexpression decreased the number of invasive cells and migration distance (Fig. 2G-H). The above results found that MT1G repressed the growth and metastasis of PTC

cells.

3.3. METTL3 mediated the m6A modification and stabilization of MT1G

By MeRIP analysis, we found that MT1G rich in m6A methylation (Fig. 3A-B). TCGA database predicted that METTL3 was lowly expressed in PTC tissues and its expression was correlated with tumor stage (Fig. 3C-D). Further analysis revealed that METTL3 expression was downregulated in PTC tissues and was positively correlated with MT1G expression (Fig. 3E-F). Furthermore, SRAMP (<http://www.cuilab.cn/sramp/>) found the presence of m6A modification sites on the MT1G mRNA (Fig. 3G). TPC-1 and SW579 cells were transfected with sh-METTL3/METTL3 overexpression vector to decrease and increase METTL3 protein expression, respectively (Fig. 3H). Through RIP analysis, we found that METTL3 knockdown decreased the enrichment of MT1G, confirming that METTL3 interacted with MT1G (Fig. 3I). In addition, METTL3 overexpression increased the luciferase activity of wild-type MT1G but had no effect on mutant MT1G luciferase activity (Fig. 3J-K). In addition, METTL3 knockdown inhibited MT1G mRNA stability, and its overexpression enhanced MT1G mRNA stability (Fig. 3L-M). Also, METTL3 knockdown reduced MT1G protein expression, and its overexpression had an opposite effect (Fig. 3N). These results suggested that METTL3 promoted MT1G mRNA stability through regulating m6A modification.

3.4. METTL3 overexpression repressed growth and metastasis of PTC cells

To confirm the METTL3 roles in PTC progression, we overexpressed METTL3 in TPC-1 and SW579 cells. METTL3 upregulation inhibited viability, colony numbers, EdU positive cell rate, and enhanced apoptosis in SW579 and TPC-1 cells (Fig. 4A-E). Besides, METTL3 upregulation reduced the invasion and migration abilities of SW579 and TPC-1 cells (Fig. 4F-G). All data indicated that METTL3 suppressed

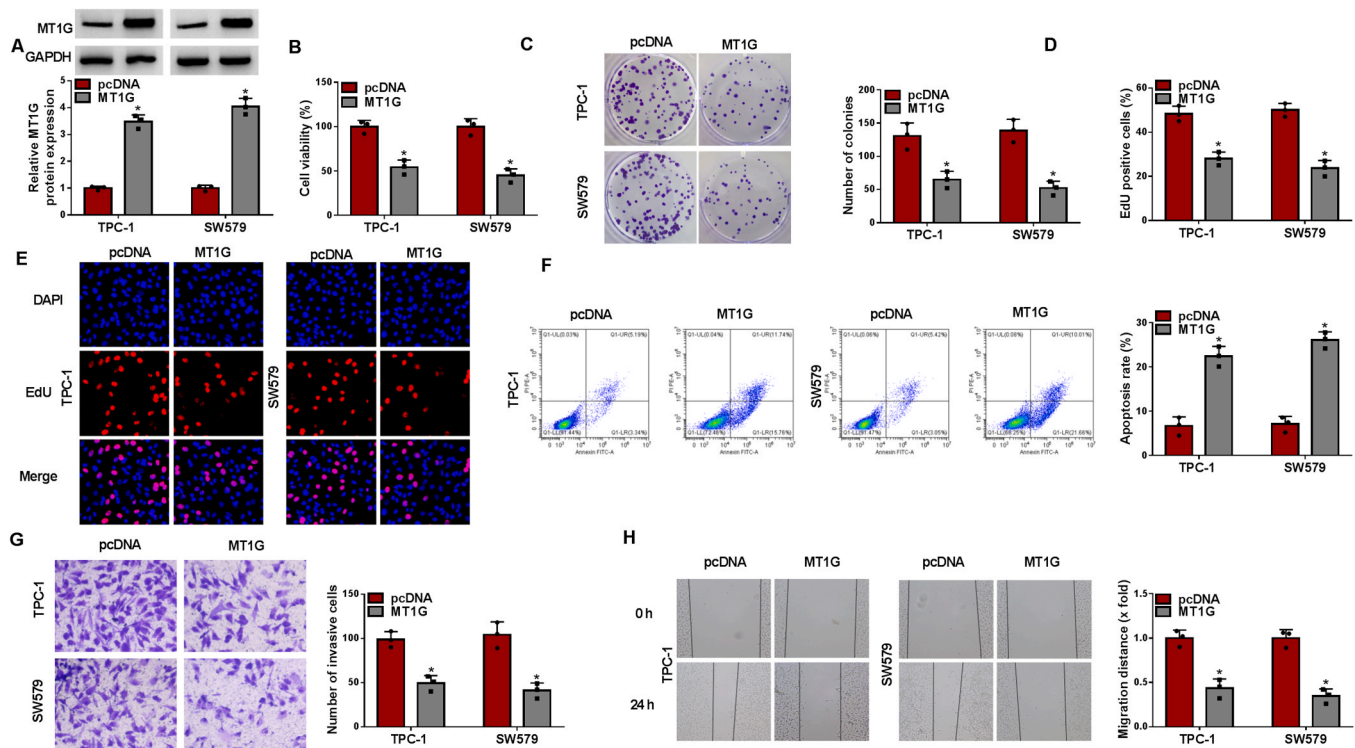


Fig. 2. The effect of MT1G overexpression on PTC cell progression. TPC-1 and SW579 cells were transfected with pcDNA/MT1G overexpression vector ($n = 3$). (A) WB analysis was used to detect the MT1G protein level. Cell proliferation, apoptosis, invasion and migration were measured by CCK8 assay (B), colony formation assay (C), EdU assay (D-E), flow cytometry (F), transwell assay (G) and wound healing assay (H). $*P < 0.05$.

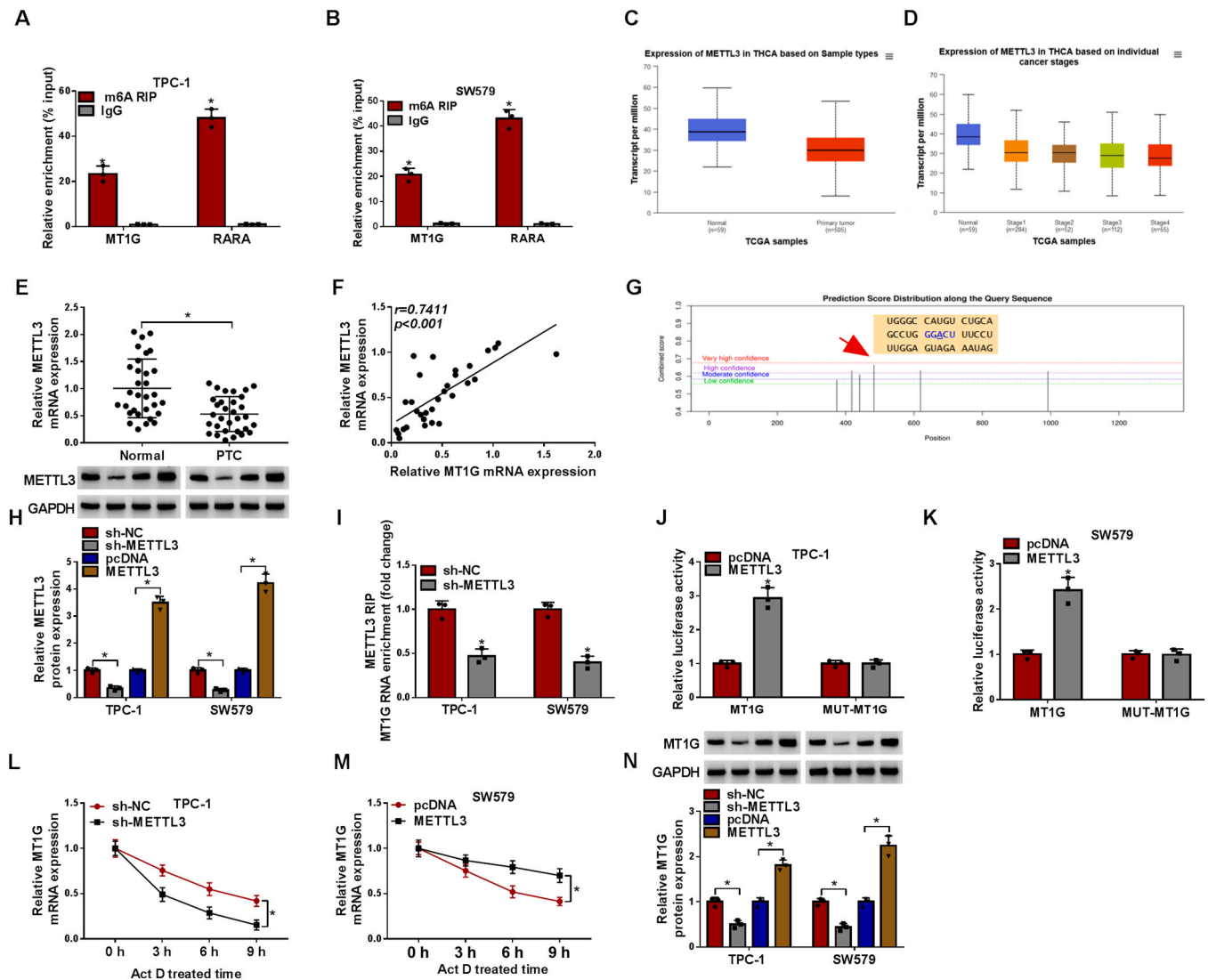


Fig. 3. Effects of METTL3 on MT1G expression. (A-B) MeRIP analysis confirmed the m6A methylation in RARA (positive control) and MT1G ($n = 3$). (C-D) METTL3 expression in PTC tissues and normal thyroid tissues was predicted by TCGA database. (E) METTL3 mRNA expression was detected by qRT-PCR in PTC tissues ($n = 31$) and normal thyroid tissues ($n = 31$). (F) The relationship between MT1G and METTL3 was assessed using the Pearson correlation coefficient. (G) SRAMP website analyzed m6A modification sites on MT1G mRNA. (H) METTL3 protein level in TPC-1 and SW579 cells transfected with sh-NC/sh-METTL3/pcDNA/METTL3 overexpression vector was tested by WB analysis ($n = 3$). METTL3 and MT1G interaction was assessed using RIP assay (I) and dual-luciferase reporter assay (J-K) ($n = 3$). (L-M) Act D treatment was used to confirm the regulation of sh-METTL3/METTL3 overexpression vector on MT1G mRNA stability ($n = 3$). (N) MT1G protein level was measured by WB analysis in TPC-1 and SW579 cells transfected with sh-NC/sh-METTL3/pcDNA/METTL3 overexpression vector ($n = 3$). * $P < 0.05$.

proliferation, invasion and migration of PTC cells.

3.5. METTL3 affected PTC process via regulating MT1G

We also examined whether METTL3 regulated MT1G to mediate the progression of PTC. MT1G protein expression was increased in TPC-1 and SW579 cells transfected with METTL3 overexpression vector, while MT1G knockdown reverted this effect (Fig. 5A). MT1G knockdown abolished the inhibition effect of METTL3 overexpression on cell viability, colony numbers and EdU positive cell rate (Fig. 5B-D). Also, METTL3 overexpression mediated the apoptosis promotion could be eliminated by sh-MT1G (Fig. 5E). Furthermore, the decreasing effects of METTL3 overexpression on the number of invasive cells and migration distance were reversed by sh-MT1G (Fig. 5F-H). Taken together, METTL3 promoted MT1G protein expression to inhibit PTC progression.

3.6. METTL3/MT1G axis regulated Wnt/ β -catenin pathway

Based on the role of the Wnt/ β -catenin pathway in PTC (Zhong et al., 2022; Geng et al., 2020), we also conducted relevant studies. METTL3 overexpression decreased the protein expression of β -catenin and Wnt3A in TPC-1 and SW579 cells, while MT1G knockdown could abolish this effect (Fig. 6A-B). Thus, we confirmed that METTL3/MT1G axis might inactivate Wnt/ β -catenin pathway to restrain PTC progression.

3.7. METTL3/MT1G axis inhibited PTC tumor growth in vivo

To determine METTL3 and MT1G roles *in vivo*, we constructed mouse xenograft model. METTL3 overexpression suppressed tumor volume and weight, and MT1G knockdown abolished this effect (Fig. 7A-B). The increase of MT1G protein expression in the tumor tissues mediated by METTL3 overexpression could be reversed by MT1G knockdown (Fig. 7C-D). Also, we observed the decreased protein expression of Wnt3A and β -catenin in METTL3 group, and this effect could be

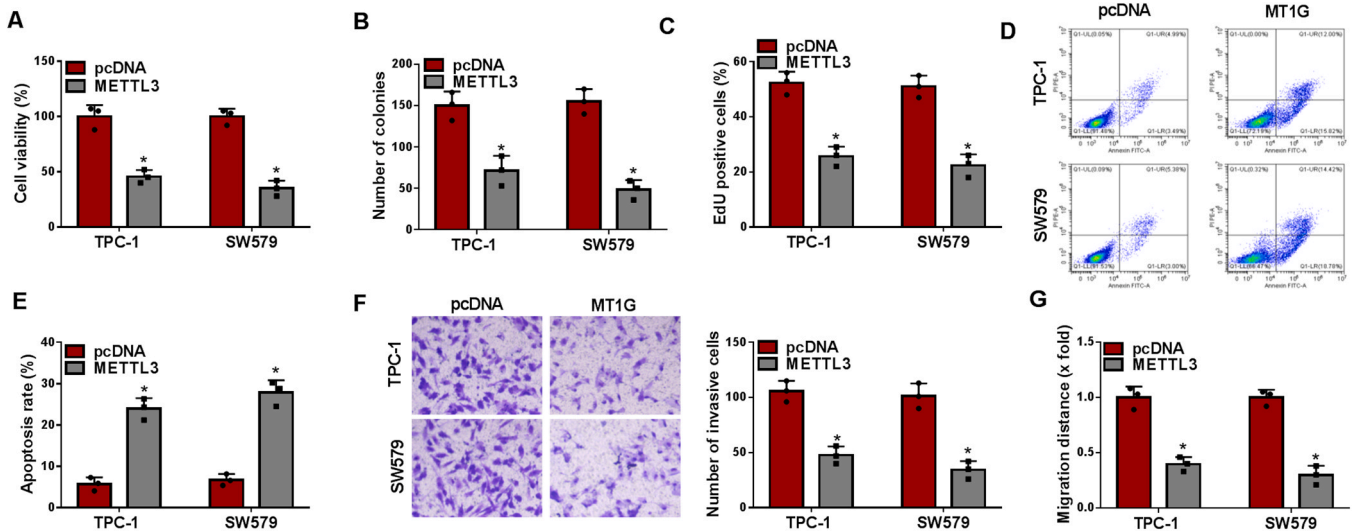


Fig. 4. Effect of METTL3 overexpression on PTC cell progression. TPC-1 and SW579 cells were transfected with pcDNA/METTL3 overexpression vector ($n = 3$). CCK8 assay (A), colony formation assay (B), EdU assay (C), flow cytometry (D-E), transwell assay (F) and wound healing assay (G) were used to test cell proliferation, apoptosis, invasion and migration. $*P < 0.05$.

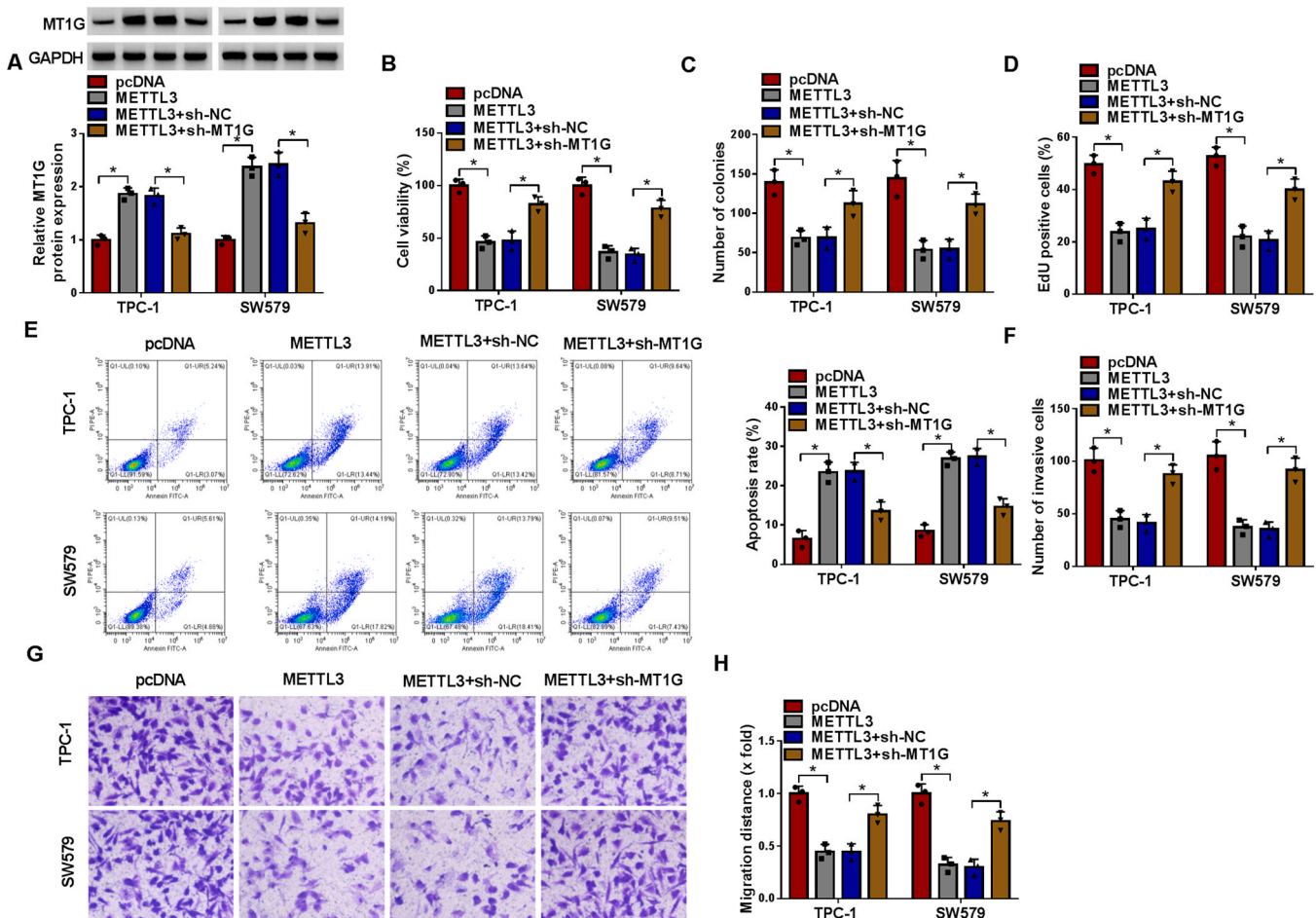


Fig. 5. Effect of METTL3 and sh-MT1G on PTC cell progression. TPC-1 and SW579 cells were co-transfected with METTL3 overexpression vector and sh-MT1G ($n = 3$). (A) MT1G protein level was tested by WB analysis. CCK8 assay (B), colony formation assay (C), EdU assay (D), flow cytometry (E), transwell assay (F) and wound healing assay (G-H) were performed to measure cell proliferation, apoptosis, invasion and migration. $*P < 0.05$.

eliminated by sh-MT1G (Fig. 7E). IHC staining showed that the Ki67 and MMP9 positive cells were reduced, while MT1G positive cells were increased in the tumor tissues of METTL3 group, but these effects were

abolished by sh-MT1G (Fig. 7F). Taken together, METTL3 repressed PTC tumor growth by MT1G/Wnt/ β -catenin pathway.

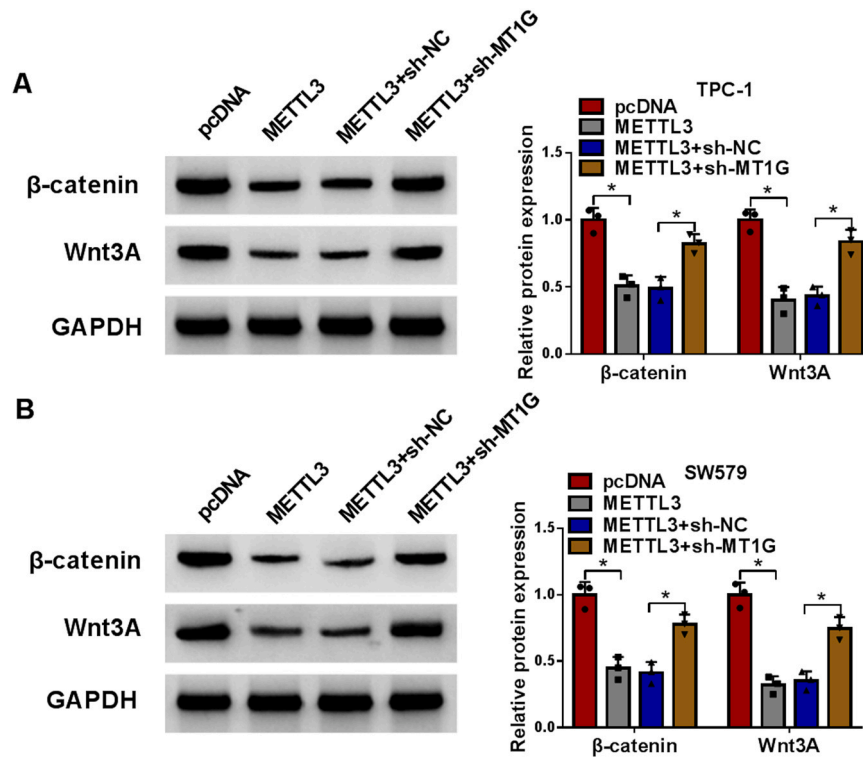


Fig. 6. Effect of METTL3/MT1G axis on Wnt/β-catenin pathway. WB analysis was used to measure the protein levels of Wnt3A and β-catenin in TPC-1 cells (A) and SW579 cells (B) co-transfected with METTL3 overexpression vector and sh-MT1G (n = 3). *P < 0.05.

4. Discussion

PTC is the most common subtype of differentiated thyroid cancers (Yang et al., 2020). A good understanding of the histological subtypes and multiple parameters of PTC is essential for PTC treatment (Lam, 2022). For example, the tall cell variant of PTC is an aggressive subtype with more prevalent high-risk features and a poor clinical prognosis (Cartwright and Fingeret, 2020). Although the treatments for PTC have been intensively explored, they still have limitations (Stan et al., 2022; Crenshaw et al., 2022). The elucidation of the underlying molecular mechanism of PTC is of great interest to patients.

Through GEO database screening, we selected MT1G to explore its role in the progression of PTC. MT1G is a highly conserved low molecular weight protein rich in cysteine residues (Krizkova et al., 2009). MT1G has been confirmed to participate in the malignant processes of cancers as a tumor suppressor. It was reported that MT1G was lowly expressed in gastric cancer tissues, and its upregulation inhibited gastric cancer cell proliferation (Xu et al., 2022). Wang et al. revealed that MT1G was downregulated in HCC tissues, and its upregulation could promote apoptosis in HCC cells (Wang et al., 2019). Although previous study confirms the low MT1G expression in PTC tissues (Ferrario et al., 2008; Wojtczak et al., 2017), the biological function of MT1G in PTC remains unclear. Consistent with previous findings (Ferrario et al., 2008; Wojtczak et al., 2017), we also detected the underexpressed MT1G in PTC tissues. Besides, MT1G upregulation inhibited PTC cell proliferation, invasion and migration, while enhanced apoptosis. These evidences demonstrated the tumor suppressor role of MT1G in PTC, similar to previous study (Ferrario et al., 2008).

Epigenetic changes occur in many physiological and pathological processes (Jiang et al., 2022). m6A is one of the most abundant RNA methylation modifications in eukaryotes (Zhang and Liu, 2022), and METTL3 is one of the m6A methylation transferase (Chen et al., 2019). Studies have indicated that METTL3-regulated m6A modification occurs in a variety of cancers (Li et al., 2022; Wei et al., 2022b). He et al. found that METTL3 was down-regulated in PTC, and its expression was

associated with PTC patients' poor prognosis (He et al., 2021). METTL3 positively regulated STEAP2 expression in an m6A-dependent manner, and it-mediated the m6A modification of STEAP2 played a critical anti-tumor role in PTC progression (Zhu et al., 2022). Furthermore, METTL3 upregulation inhibited invasion, migration, and viability of PTC cells by mediating LINC01125 methylation (He et al., 2023). In this study, we found that MT1G had methylation modification sites and its expression was positively correlated with METTL3 expression in PTC tissues. Further studies revealed that METTL3 could promote MT1G mRNA stability, suggesting that METTL3 facilitated MT1G stabilization through m6A modification. Also, MT1G knockdown reversed METTL3-mediated the inhibition on PTC cell growth and metastasis, confirming that METTL3 restrained PTC progression by upregulating MT1G. The activity of Wnt/β-catenin pathway is considered to be related to PTC progression (Zhong et al., 2022; Geng et al., 2020). Here, we measured Wnt3A and β-catenin protein levels, and found that METTL3/MT1G axis could inactivate Wnt/β-catenin pathway. *In vivo*, METTL3 suppressed PTC tumor growth via inactivating Wnt/β-catenin pathway by enhancing MT1G expression. These data provide evidence for the existence of the METTL3/MT1G/Wnt/β-catenin axis in PTC. Despite the insights gained into the METTL3-mediated m6A modification of MT1G in suppressing PTC progression via the Wnt/β-catenin pathway, future studies would benefit from RNA sequencing (RNA-seq) analysis following MT1G expression modulation. Such an approach could facilitate the identification of additional potential targets and pathways regulated by METTL3.

In conclusion, we mechanistically elucidated the anti-PTC role of MT1G. Our research showed that METTL3 could promote MT1G mRNA stability by methylation modification, thereby inactivating Wnt/β-catenin pathway to suppress PTC cell growth and metastasis. The discovery of METTL3/MT1G/Wnt/β-catenin pathway provides new ideas for developing the therapeutic options of PTC.

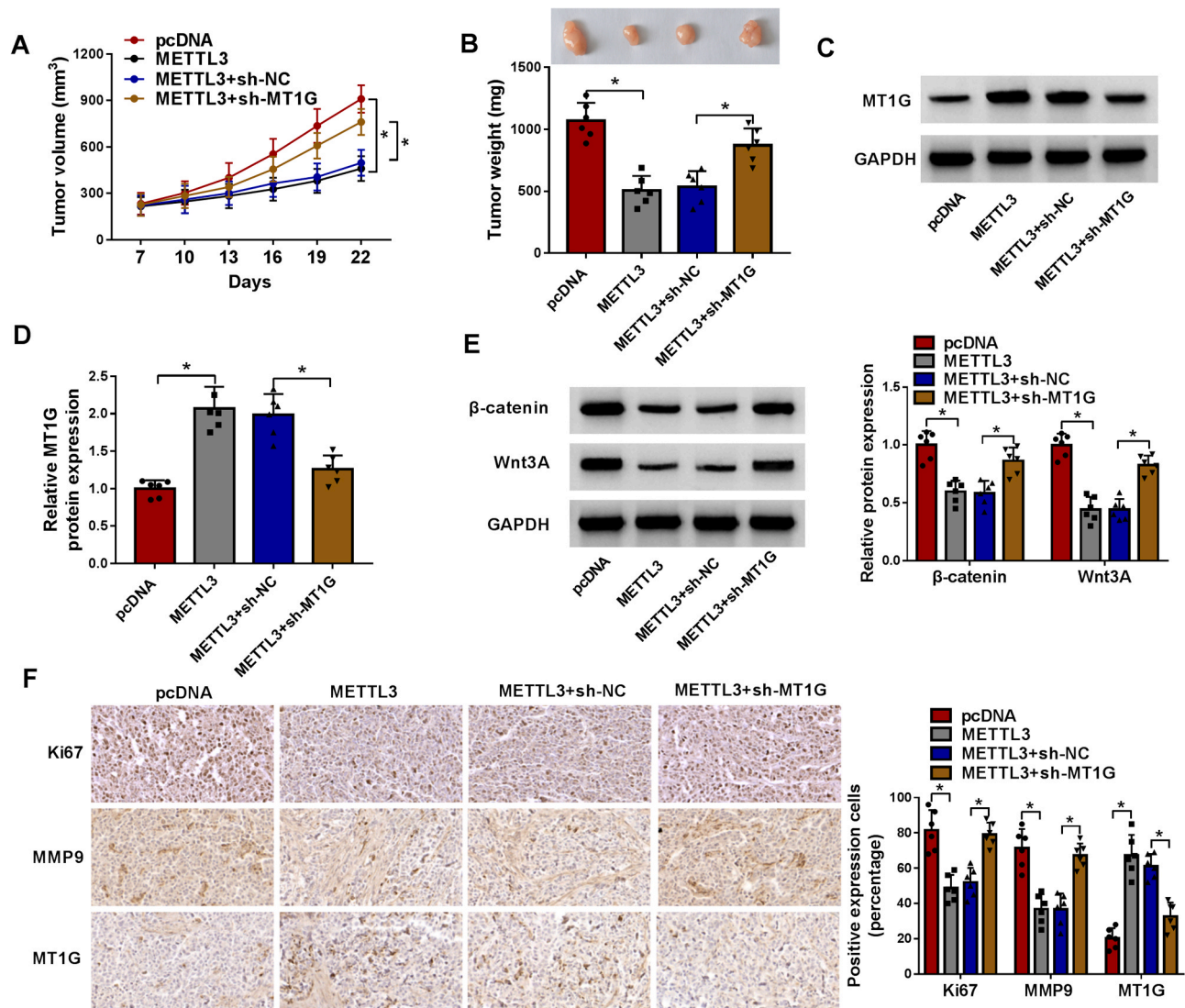


Fig. 7. Effects of METTL3 and MT1G *in vivo*. (A-B) Tumor volume and weight were measured (n = 6). (C-E) The protein levels of MT1G, Wnt3A and β-catenin in mice tumor tissues were detected WB analysis (n = 6). (F) Ki67, MMP9 and MT1G positive cells in mice tumor tissues were measured by IHC staining (n = 6). *P < 0.05.

Ethical Approval

This research had been approved by the Ethics Committee of Cangzhou Hospital of Integrated TCM-WM (2021-KY-057.1) and all operation procedures comply with relevant regulations.

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Not applicable.

CRediT authorship contribution statement

Liu Jie: Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Investigation, Conceptualization. **Wang Kun:** Writing – review & editing, Methodology, Investigation. **Wang Xueyu:** Writing – review & editing, Visualization, Formal analysis. **Chen Yating:** Writing – review & editing, Visualization, Investigation. **Yao Lei:** Writing – review & editing, Methodology, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no conflicts of interest

Acknowledgment

None

Consent to Participate

Written informed consent was obtained from all participants under the relevant guidelines and regulations.

Consent to Publication

Written consent from the patients was obtained for publishing their clinical information.

Data availability

Data will be made available on request.

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